The University of New South Wales
**ELEC3115 – Electromagnetic Engineering**

# Experiment E: Eddy currents, hysteresis & core losses

## Part I Eddy currents

### 1.1 Objectives

1. To demonstrate eddy currents
2. To observe the beneficial effect of using laminated iron cores
3. To be familiar with the theory of hysteresis loops
4. To display a hysteresis loop on the oscilloscope
5. To measure the core losses in a laminated core
6. To measure the core losses in a solid steel core

### 1.2 Brief theory

**Eddy currents and laminated cores**

An alternating magnetic field is always surrounded by an alternating electric field, as shown in Fig 1.

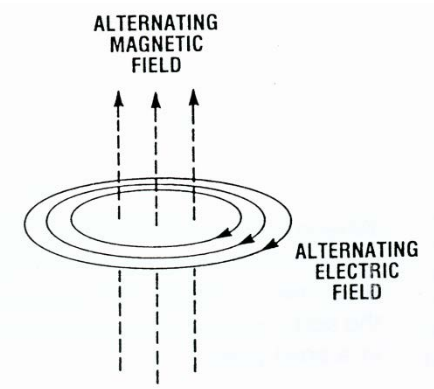


Fig 1: Relationship between magnetic and electric field

If an alternating magnetic field passes through a metallic conducting body, the alternating electric field will cause the free electrons in the body to move back and forth inside the body, causing an alternating current flow in the body. This alternating current flowing in the body will result in power loss and temperature rise of the body. This power loss is usually undesirable, except for in induction cooking ovens where such loss is actually desirable. The path of the current is circular, and it flows back and forth throughout the whole body. This current is known as ***Eddy*** or ***Foucault current***.

Whenever an alternating magnetic flux occurs in a conducting body, it produces eddy currents and power losses within the material. One way to reduce these losses is to construct the body with a number of thin laminate sheets. If these laminations are insulated from each other, the eddy currents are confined to smaller areas, and so they are much smaller than in an un-laminated body.

The power loss in the material due to eddy currents caused by a sinusoidally time-varying magnetic field is given as,

$$P_e = K_e f^2 B_m^2 \tau \quad \text{W/m}^3$$

where $K_e$ is a constant related to the conductivity of the material, $f$ is the frequency in Hz of the field variation, $B_m$ is the peak magnetic field in the material and $t$ is the thickness of the laminations, all in SI units.

### Hysteresis loss and the BH loop

When a magnetic material is subjected to a cyclic magnetic field at a frequency $f$ Hz, the magnetic dipole orientations in every cycle are accompanied by friction within the material, resulting in *Hysteresis* power loss. This loss in the magnetic material per unit volume is given by

$$P_h = K_h f B_m^n \quad \text{W/m}^3$$

where, $n$ = Steinmetz constant = 1.6 – 2, $K_h$ is a constant, $f$ is the frequency of field variation, and $B_m$ is the peak field in the material.

$P_h$ is known as the hysteresis loss (a type of power loss) in the magnetic circuit. The constants $K_h$ and $n$ are determined by the shape of the BH curve and other physical properties of the material of the core. A typical BH loop of a magnetic material is shown in figure 2. The energy loss/unit volume during each cyclic magnetization is given by the area within the BH loop.

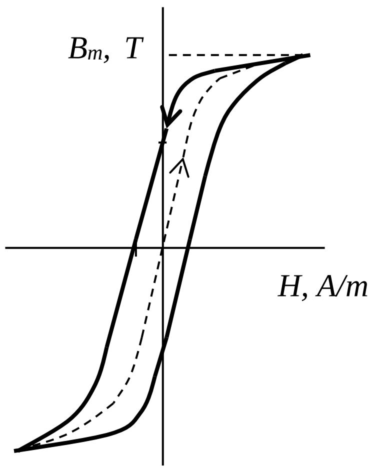


Fig. 2

These two losses are often grouped together and called *Iron* or *Core* losses. This is because these losses mostly take place in the iron core of a magnetic circuit. Thus,

$$P_c = K_h f B_m^n + K_e f^2 B_m^2 \quad \text{W/m}^3$$

**Required Equipment**

| | | |
|---|---|---|
| AC Ammeter | AC voltmeter | Connection leads |
| Coil | AC Power supply (Variac) | |
| Mounting base | Aluminium ring (indicated as ⑱in figures) | |

3 x 133 mm Laminated bars (indicated as ① in figures).
133 mm laminated bar with hook (indicated as ③ in figures),
178 mm laminated bar (indicated as ④ in figures).
133 mm steel bar (indicated as ⑥in figures)
Oscilloscope
A thermometer

**1.3 Procedure:**

## Part I: Measurements of eddy current loss

**CAUTION**
**Do not make any connection or change to the circuit when the power supply is on. You may need to use an appropriate scale of ammeter and voltmeter. While changing from one scale to another in any meter, turn off the power supply first. Do not disconnect anything while power is on.**

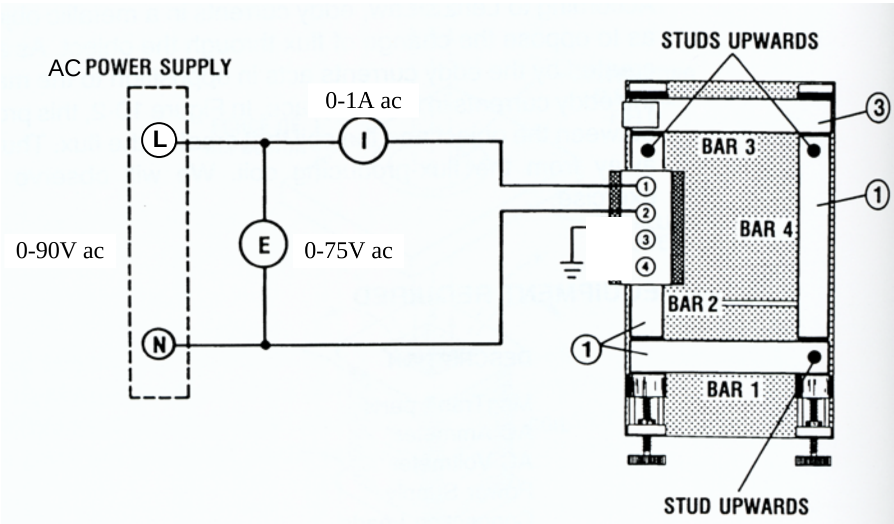


Fig 3: circuit for observing the effects of a laminated construction on electromagnetic induction

1. Set up the circuit of Fig 3. Tighten the screws to minimize the air-gaps in the magnetic circuit.
2. Turn on the power and adjust voltage E to 50 V ac. You will keep the circuit in operation for 5min. Record the input voltage, current and temperatures of the four bars in a **Table** at the start and end of 5 minutes.
   Turn off the power.

**Question 1**

What was the reason for the temperature rise in the bars? (record your answer in your logbook)

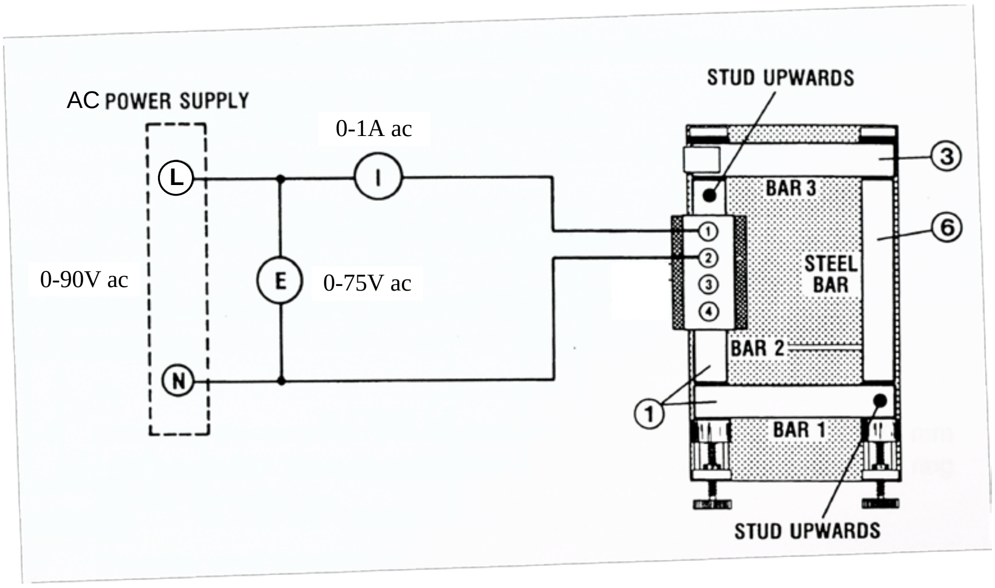


Fig. 4 Circuit for observing the effect of a solid steel bar on electromagnetic induction

3. Set up the circuit of Fig 4 using a 133 mm solid soft steel bars instead of a laminated bar. Tighten the screws.

4. Turn on the power and adjust voltage E to 50 V ac. Measure and record current *I* and keep the circuit in operation for 5min. Record the input voltage, current and temperatures of the four bars in a **Table** at the start and end of 5 minutes.

   Turn off the power.

**Question 2**

Explain the difference in temperature rise for the two types of bars. (Record your answer in your logbook.)

5. Set up the circuit of Fig. 5 in which the two bars are mounted vertically. Tighten the screws. Make sure terminal 1 of the coil is nearest the left bar, as shown.

6. Turn on the power and adjust voltage E to 30 V AC. Record the value of *I*. **Current *I* should not be allowed to exceed 0.9 A**.

7. Place the aluminium ring over the right bar and observe its position.

| **CAUTION**<br>**Aluminium ring can become very hot!** |
|---|

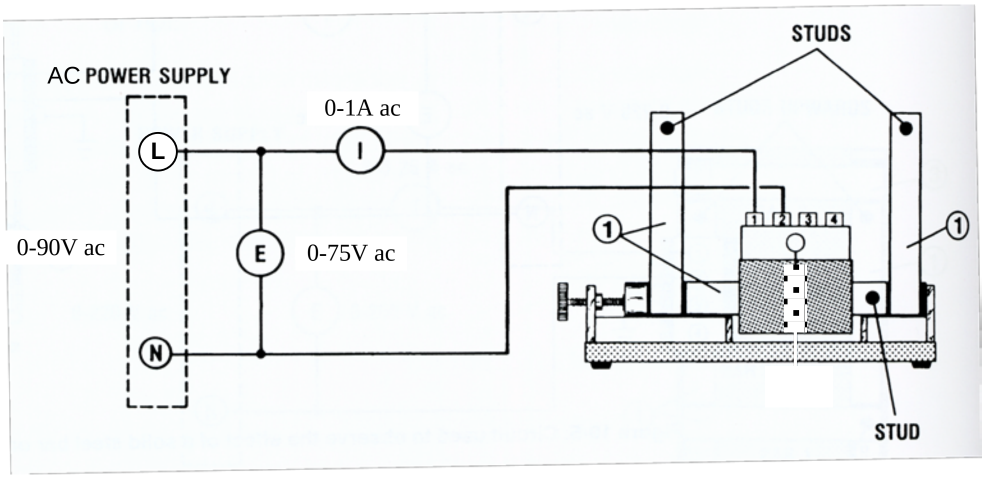


Fig. 5 Circuit used to observe the effect of eddy current in a metallic ring

Let the ring float for about 1 min and observe that it becomes quite hot. Record this temperature.

8. Measure and record the value of *I* with and without the ring while keeping *E* at 30 V ac.

**Question 3**

Record your observation of the temperature and height of the aluminium ring on the vertical bar.

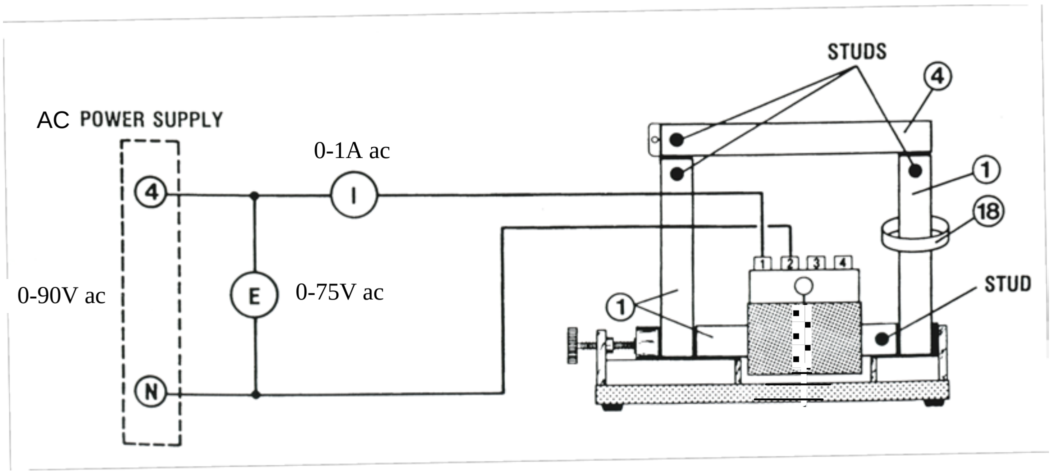


Fig. 6 Circuit used to observe the effect of eddy current in a metallic ring

9. Complete the magnetic circuit, as shown in Fig**.** 6, using a 178 mm laminated bar to connect the upper end of the vertical bars, and observe the height of the aluminium ring around the right bar when current is increased slowly to 0.9 A ac for a short time.

10. Replace the bar that holds the ring with a solid (unlaminated) bar and observe the relative height of the aluminium ring when the current is increased slowly as above.

**CAUTION**
**Do not exceed current above 0.9 A.**
**The ring will become very hot: handle with care!**

**Question 4**

Comment on the temperature rise and the relative height of the ring around the vertical bar. Explain the differences in heights of the ring and temperature rise with laminated and unlaminated bars.

11. Switch off the variac, by first bringing the voltage to zero.

## PART II: The BH loop, measurements of hysteresis and core losses

The BH loop of Fig.2 may also be drawn as in Fig. 7, in which fluxes and currents replace B and H. When a coil, with an iron core and having N turns, is connected to an AC source, it draws an AC current *I*, an alternating magnetic flux $\phi$ is produced in the iron core. If the instantaneous flux $\phi$ is plotted against the instantaneous current I, we obtain a closed-loop whose general shape is as shown in Fig 7, similar to the BH loop of Fig.2. Normally a hysteresis loop represents the variation of the magnetic flux density B with the magnetic field intensity H.

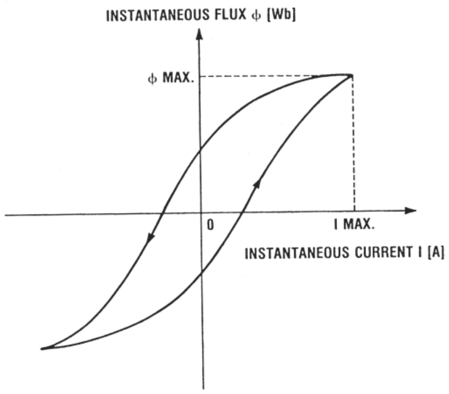


Fig.7 Hysteresis loop

Note that the total power loss at the input measured at the input of the coil exciting the core is composed of both hysteresis loss and eddy current losses, in the iron core, if the small $I^2R$ loss in the copper wiring of the coil is neglected. It can be shown that area of the loop (in Wb-A) is proportional to the core loss in Joules per cycle of magnetization of the coil. Therefore, total hysteresis loss $P_h$ is given by

$$P_L = ANf \qquad (1)$$

where, $P_L$ the total loss in W, $A$ the area of a hysteresis loop in Wb-A, $N$ is the number of turns on the coil that produces the flux, and $f$ the frequency of the source in Hz.

The hysteresis loop can be displayed on the screen of an oscilloscope as an XY plot if instantaneous current $i$ and flux $\phi$ are first converted to equivalent instantaneous voltages. This can be done using the current probe and the flux meter, but not with the CROs provided. Without such a CRO, you will need to measure the voltages and plot the curve by hand. To use a current probe, open the clamp and place it around the wire whose current is to be measured. In the current probe, the current converts into a proportional voltage $v_1$ by placing a resistance R in series with the winding. The value of $R$ is sufficiently small so that the $IR$ drop is negligible compared with supply voltage $E$ (as shown in Fig. 8). A search coil $S$ is used with the flux meter to sample the flux, and a specialised integrating circuit T in the flux meter makes the output voltage $V_2$ proportional to the instantaneous flux.

Note: In the hysteresis loop obtained in this experiment, the intercept on the horizontal axis is not a measure of the coercive force ($H_c$), and the intercept on the vertical axis is not a measure of remanent magnetism ($B_r$), as in regular hysteresis loop.

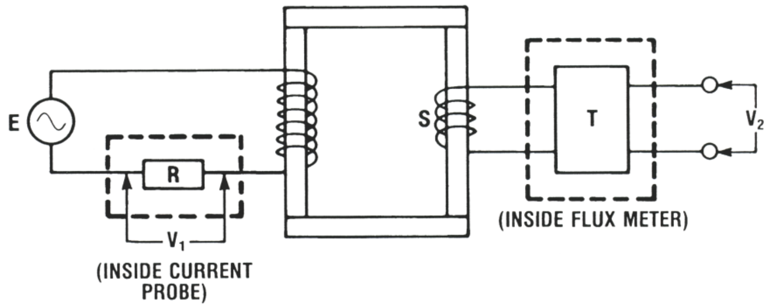


Fig 8: Using the current probe and the flux-meter

**Extra equipment required**
BNC/BNC cables
Oscilloscope
Current probe

**Procedure**

| **CAUTION** |
|---|
| **Do not make any connection or change to the circuit when the power supply is on.** |

**The BH loop on the CRO**

1. Set up the circuit with a digital power meter, as shown in Fig. 9. The power meter will measure the *total* power input to the circuit. Tighten the screws to minimize air-gaps. Set the oscilloscope to DC input mode. Select the 0-1000 μWb peak scale on the flux meter.

2. Apply power and raise voltage E gradually to 55 V AC. Observe the display on the oscilloscope. Note that the theoretical values of $\phi_{max}$ is found using the formula,

$$\emptyset_{max} = \frac{E}{4.44fN}$$

3. Adjust voltage $E$ in steps of 5V and for each value of $E$, record both $V_{1max}$ (CH1 on the oscilloscope) and $V_{2max}$ (CH2), $E$ and $I$. Then calculate the corresponding maximum values of current $I$ and flux $\phi$. Note the sensitivity of the current probe is 100 mV/A and the sensitivity of the flux meter is 1 mV/μWb. Therefore,

$I_{max}(A) = V_{1max}(mV)/100$
$\phi_{max}\ (\mu Wb) = V_{2max}\ mV.$

4. To observe the hysteresis loop in the oscilloscope screen, select XY display mode on CRO.

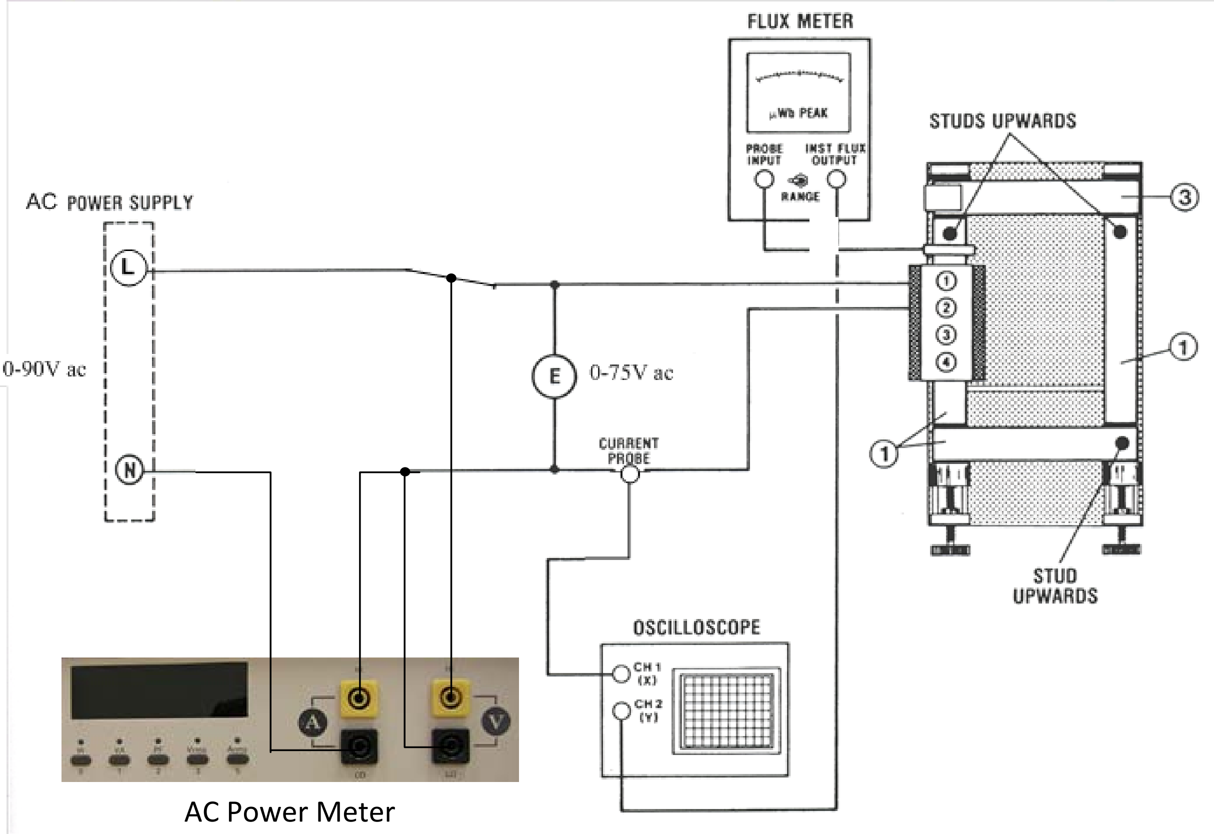


Fig. 9. Circuit used to observe the hysteresis loop

<u>Procedure to select X-Y mode:</u>

Press ‘Acquire’ button in the oscilloscope. It will bring a screen menu. Select “XY” of the screen menu by pressing the corresponding button at the bottom of the screen. Then select “XY triggering” from the vertical menu by pressing the corresponding button at the right-hand edge of the screen. You can adjust the scales of X and Y by adjusting relevant channels.

**Question 5**

Comment on your observation of how the hysteresis loop changes shape as voltage *E* is varied. Note that its area increases with increasing voltage.

**Note: The BH loop on the CRO may be quite thin when E is small, and you may find it difficult to plot these in your logbook. Try to do your best.**

5. Plot the hysteresis loop for a few values of *E* in your logbook using the graph paper at the back of this lab sheet.

6. Using a short lead, loop it around the bar without a coil and short-circuit the lead itself. In this case, the hysteresis loop becomes wider. Explain why.
   Turn off the power.

7. **Graphical measurement of the hysteresis loss:**

   Because of the thin hysteresis loops for low values of $E$, calculate the hysteresis loss only for highest $E$ = 55V. Remember that this loss/cycle of magnetization is the area contained within the loop.

8. Calculate the area of the loop in units of $V^2$ using the following procedure from the plot of your hysteresis loop.

   a. Number of square divisions in the loop = ---------- (Count the number of small squares inside the loop and divide by 25*)
   *small square division contains 25 smaller square divisions.

   b. Area of each square division = X-scale × Y-scale

   =----------V/Div × ----------V/Div

   =---------- $V^2/Div^2$

   c. Area of loop = number of square divisions × area of each square division

   = ------------- $V^2$

   Calculate the area $A$ of the loop in Weber-Amperes:

   $A$ = Area of loop in $V^2$ / (0.1 V/A × 1000 V/Wb) = ------------- $V^2$ × 0.01Wb A/$V^2$

   = -------------Wb A

   Number of the turns on coil A, N = -------------

   Frequency of the source $f$ = 50 Hz

   The hysteresis loss is given by, $P_h = ANf$ = ------------- W

9. For the highest input voltage $E$ and current $I$, calculate the $I^2R$ loss in the coil using the RMS current $I$, and the resistance of the coil. Given:

   Resistances of the coils are: Coil A: ------------- Ω for 2400 turns, ------------- Ω for 600 turns;

   Coil B: -------------Ω for 2400 turns, ------------- Ω for 600 turns)

10. Calculate the eddy current loss by subtracting from the total input power, the $I^2R$ and the hysteresis losses.

**Question 6**

Comment on the three types of power losses in the coil as fractions of the total input power from the AC source.

11. Replace the right laminated bar without a coil in Fig. 9 with the 133mm solid, soft steel bar. Apply power and raise voltage $E$ to 55V ( the highest voltage you used above). Observe the display on the oscilloscope. Measure $V_1$ max and $V_2$ max and calculate the corresponding maximum values of $I$ max and flux $\phi$.

$V_{1max}$ = ____________V $I_{max}$=_____________ A

$V_{2max}$ =_________________V $\phi_{max}$ = ______________μWb.

12. Plot the hysteresis loop in your logbook.

13. Calculate the core loss using the same procedure as in step 8 but using now new hysteresis loop. First Calculate the area of the loop in $V^2$.

 a. Number of square divisions in the loop = ------------- ( Count the number of small squares inside the loop and divide by 25*)

 *small square division contains 25 smaller square divisions.

 b. Area of loop = number of square divisions × area of each square division

 = -------------$V^2$

 Calculate the area $A$ of the loop in Weber-Amperes:

 A = Area of loop in $V^2$ / (0.1V/A × 1000 V/Wb) =-------------$V^2$ × 0.01 Wb. A/$V^2$

 = -------------Wb. A

 Number of the turns on coil A , $N$ = -------------

 Frequency of the source $f$ =50 Hz

 Calculate the hysteresis loss using , $P_h = ANf$ =------------- W

14. Repeat step 9 and 10, in order to separately find the three types of losses in coil for this case.

Turn off the power. Disconnect the circuit and return all cables to their holders.

**Question 7**

Comment on the changes in the hysteresis and the other two types of power losses for this circuit. Explain why.